CHAPTER 1

EVOLUTION





Figure-1: Different types of Dinosaurs

There are hundreds of lakhs of species present on this earth. We have observed some of them while we have heard of some others. There are several others that exist no more now that is, they have become extinct. Today fossil remains are evidences of their existence long ago. "Dinosaurs" are some of the groups of extinct species. You all may have heard about dinosaurs. Their fossils have been found at different places on earth. Even in India their fossils have been found from time to time at several places in the Narmada valley like "Lametaghat" near Jabalpur. Now we know that these regions in the Narmada valley were habitat of dinosaurs'. Fossils have helped us to know that dinosaurs were of different types and present on earth in large numbers from around 2000 to 600 lakh years ago. The size of dinosaurs ranged from a minimum of nearly 50 cm to a maximum of around 40 metres. They were of several types; some carnivores, some herbivores, some bipeds (walking on two legs), some quadrupeds (walking on four legs) while some could even fly. Diverse forms of flora and fauna were present during the time of dinosaurs. Several plants similar to those that are found nowadays like algae, fern, moss, cycads, gingko and animals like mammals resembling rats, insects such as cockroaches etc. were present during the time of dinosaurs. The birds that we see now are considered as small forms of dinosaurs. This list may extend even further. By the time of extinction of dinosaurs a number of giant mammals like elephants were observed to inhabit areas where the dinosaurs lived.

Dinosaurs are no longer seen today but certain species of plants and animals similar to those of the age of dinosaurs are still observed today.

- Think how this may have happened?

Do You Know?

This is a species of gingko tree similar to that found during the period of dinosaurs. This was the only form of life that could withstand the atomic-bomb blast in Hiroshima.



1.1 Habitat and its effect

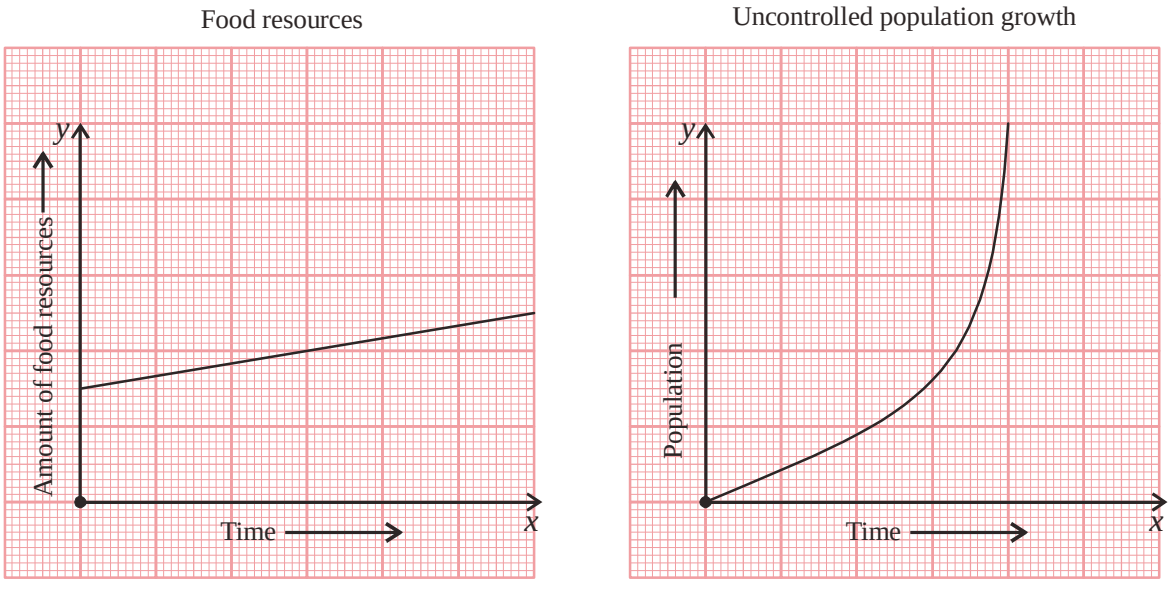
You have studied in the chapter on habitat in class 9 that, habitat is a place where the basic needs (food, reproduction and safety) of any living organism is fulfilled.

- When dinosaurs were abundant on earth, do you think there were enough resources in the habitat for their survival?
- Would their population increase if resources were adequate?

We observe presently, that the population of some species is increasing at a fast rate while that of some others is declining rapidly. As for example, the number of tigers in our country are reducing at such a fast pace that they are at the edge of extinction. The same thing is happening with vultures, sparrow and certain other animals. On the other hand human population is increasing at such a fast pace that there is a rapid fall in the availability of resources in human habitats.

- You have read about the food chain. What will happen if any organism of a food chain vanishes?
- What do you think may be the cause for the rapid decline in the number of tigers? Does the increase in human population have any role?
- What will happen if the population of a certain species of organism increases unusually?
- Is the rate of population growth of a species similar to the rate of the growth of food and other abiotic resources?

We find that indeed there are certain factors that may control increasing number of organisms. We shall take the help of the graphs to study one of them. A study conducted way back in around 1780 claimed a situation as showed in the following graphs. Here one graph shows the uncontrolled increase in population size while the other shows amount of available food resources for organisms of a species.



2 (a) 2 (b)

Figure-2: Graph showing increase in food resources and uncontrolled population growth

Study the graphs and answer the following-

- Is the rate of population growth and that of food resource same?
- Which graph shows growth at a faster rate?

- Would the amount of food resources have any effect on the rate of growth of population?

The graph shows that if population of a species grows uncontrollably then the food resources will not be able to fulfil the requirement of the population. In this situation, control in number due to natural disasters or deaths due to conflicts related to competition for resources may occur.

- Do organisms that survive have some special abilities?
- Does nature have any influence on the number of organisms?
- Could declining amount of resources in habitat be a cause for extinction of dinosaurs?

Scientists have been wondering about such questions on extinction, origin of species their diversification etc. Questions like; what is the process of origin of organisms? Were all organisms in their present form placed on earth from somewhere as such? Or do organisms that survive on earth change over time? Do organisms become extinct as new forms take their place?

You have studied previously that new cells originate from pre-existing cells. The pre-existing cells originally being formed from certain compounds. Also that, cells organise to tissues, tissues to organ, organs to organ systems to multicellular forms of organisms. Are organisms formed from pre-existing organisms just as cells?

A little over 200 years back eminent naturalists(scientists studying nature) Charles Darwin and Alfred Russell Wallace were looking for answers to such questions, there were of course many before them as well who tried to answer such questions in their own way. But these two made an extensive study of organisms, their remains, natural conditions and proposed the most profound theory in science that brought about a remarkable change in the views of the society till then. Let us study mainly about Darwin and his findings to see how an important theory was arrived at.

1.2 Voyage of the Beagle and Darwin's Experience

Since childhood Darwin was interested in studying and gathering information about natural phenomenon, habitats of different organisms and their behavior. He studied about diversity of living organisms and the influence of nature on the diversity. His observations completely changed the social and scientific thought of his time.

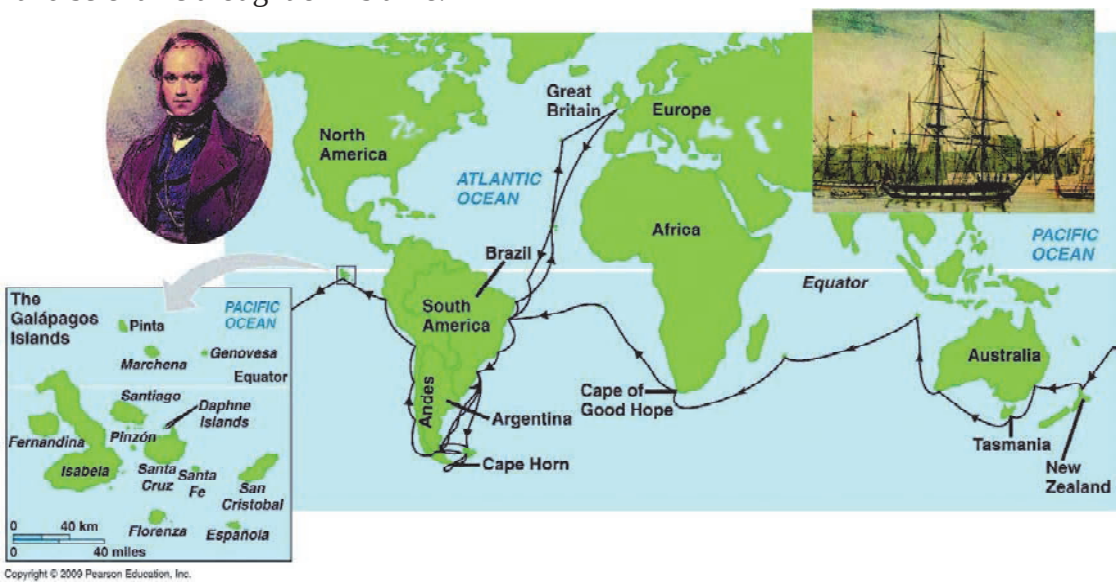


Figure-3: Map showing the areas surveyed during the voyage by HMS Beagle

In the year 1831, Darwin got an opportunity to travel around the world on a survey expedition by the ship, HMS Beagle. He was to give company to the captain of the ship. The Beagle expedition was undertaken to map different regions of the world, explore mineral resources, study biodiversity etc. During the voyage, that started from England and went via South America, Darwin found a chance to roam around different islands of the world. It was the first time that he studied organisms of tropical rain-forests. He collected and observed different organisms as well as remains of organisms of the distant past. Apart from the accommodation and study room there was a good library in the ship. The voyage started on 27th December 1831 and ended on 2nd October 1836.

1.3 Darwin's Observations

Some of the major observations during the voyage, which helped Darwin to propose the theory of evolution, are as follows-

- Darwin studied samples of sea water and was surprised to see various types of organisms in those samples. Here he observed a beautiful chain of food and feeding among even the microscopic forms of life! The forests and mountains of South America had several such examples too. Everywhere he found food chains and food webs where an organism (predator) fed on the other(prey). These predator prey relationships made him wonder- "There is struggle for life everywhere, either kill or die. Those able to survive, live on. Perhaps those that get accustomed to geographical conditions survived!"



Figure-4: Fossil collected by Darwin

- He found fossils of aquatic organisms and that may have lived in the sea earlier on the mountains of South America. During his studies there, he experienced an earthquake as well. The ocean level rose to nearly 3 meters due to the earthquake. These types of observations were indicating that sudden severe changes may occur on islands or in the oceans over time.
- Darwin found that slow and continuous changes on earth could also bring about changes in land forms on earth. Beaches surrounded by cliffs rather than plain sandy stretches indicated this.
- The fossils of giant mammals, reptiles etc. showed similarities as well as differences as compared with the existing organisms. Thus, as the surface of the earth changed over time, so might have the organisms.

Do you know?



Fishermen caught a peculiar fish along the sea coast of South Africa in 1938 that caused a stir among the scientific community. An organism that was thought to have become extinct along with the dinosaurs was found! It was believed to have undergone no changes in features which were similar to those of the period of dinosaurs and even before that. It was thus called as a living fossil. It had certain features like reptiles and certain others like fishes. This was a promising missing link between reptiles and fishes. We now know that its population has evolved over time and is still undergoing evolution so it may not be right to still call it as a living fossil.

1.4 Some important observations of Darwin on Galapagos islands

Darwin collected fossils and several specimens of organisms that he found on Galapagos.

- What do you think fossils help us to study about?

The speeimens and fossils helped him to study and compare forms of present organisms with those of the past. As for example, he observed that armadillo of the past were larger in size than those of the present (see Figure- 6). "Armadillos had changed over time", thought Darwin.



Figure-5: Different organisms observed by Darwin in Galapagos Islands - Monitor lizards(60-100 cm) duck like Boobie birds(70-90cm) Cactus(600-700cm) Finch(10-20cm) Tortoise (120-150cm)

Darwin had collected fossils of molluscs (like snails), reptiles, mammals etc. as well as specimens of the existing forms. Fossils are important as they often provide evidences of mutual relationships between organisms of the past and the present as well as between organisms present now.

While studying about the finches of the Galapagos Islands, Darwin was simply amazed to see the variations among them. Finches are birds similar to sparrows of our country. Darwin had studied nearly 15 finch species. As he observed finches of different islands as well as the mainland, he found certain variations in spite of basic similarities among them. The variations were most prominent in the type of their beaks and coat colour. He measured the dimension of the beaks and found them to differ a lot. Birds with different beaks, he observed, had different food habits as well. While some birds ate seeds, some others ate fruits. Some of the finches ate insects and worms while others survived on the nectar of flowers growing on the island. Variation in beaks appeared to correspond to the variation in the types of available food resources. So, Darwin remarked that structural variations in beak were related to their functional variations.

He was of the opinion that finches had originally flown from Ecuador(on the mainland, refer to figure-3) to the different islands of Galapagos. Groups of finches started thriving in different environments of these islands. A group most suited to the environment evolved most and could be seen to have characters conducive for their survival on these islands. This could only have been possible due to variations in the group of finches.

The development of different communities of finches surviving on different islands of Galapagos from a community that had originally come from Ecuador showed that varied finches originated from an apparently similar group. Finches with prominently visible variations in type of beak were related and had the same origin. Such characters as the type of beak in finches,which had diversified structurally and functionally from the same original type, are called as,'homologous characters'. Darwin had considered these as very strong evidences for evolution.

Let us observe some characters first in plants and then in animals to find out more about homologous characters.

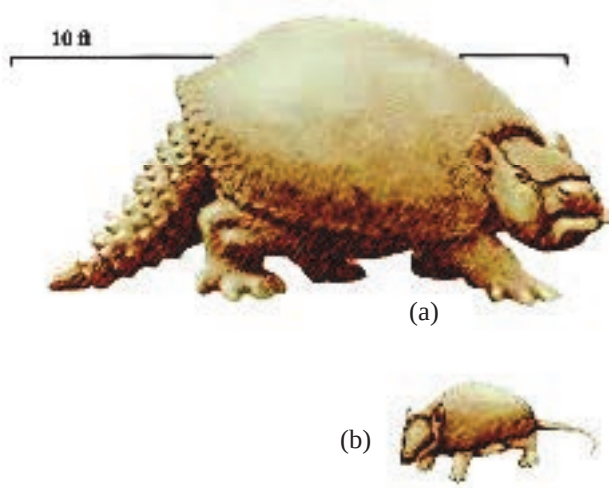


Figure-6: Ancient and present Armadillo

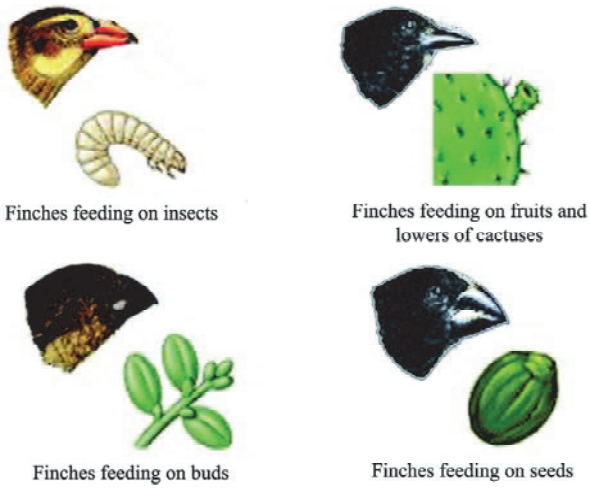


Figure-7: Different types of beaks of finches

Activity-1: Observation of homologous characters

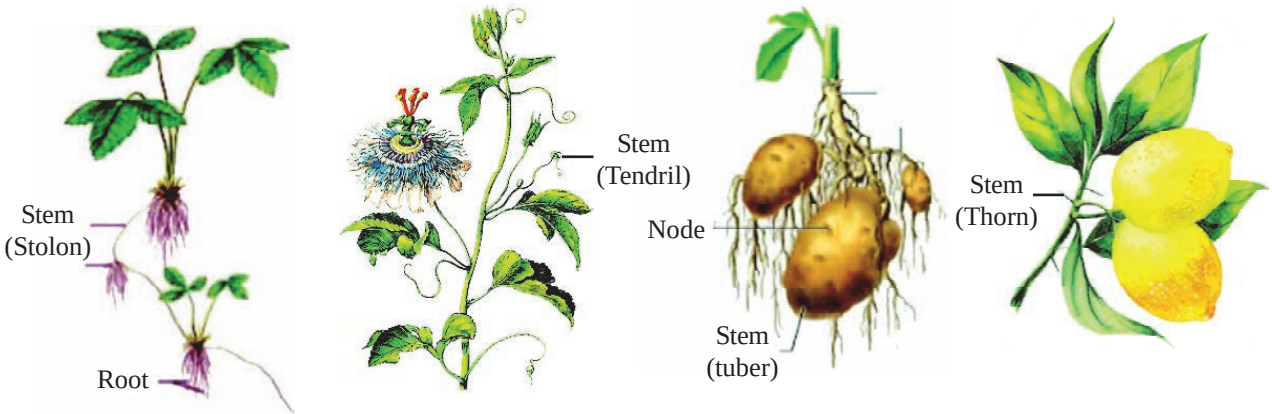


Figure-8(a): Some homologous characters in plants

You may have observed different types of stems in plants. Some types are shown in figure 8a. With the help of the figure, your own prior observation and discussion with your friends enlist all the different plants in which you can see stems that appear to perform different functions. Write the functions that you believe the stems must be having. Take the help of your teacher or other books to check your answer. Figure 8a shows stems of strawberry that have role in reproduction(called as stolon here), stems of passion flower that have role in climbing (stem tendrils), potato stems (tuber) that have mainly the function of storage of food and reproduction and stem of lemon developed into thorns (mainly protective in function. All these stems have diversified both structurally and functionally but indicate the same origin as all these are stems.

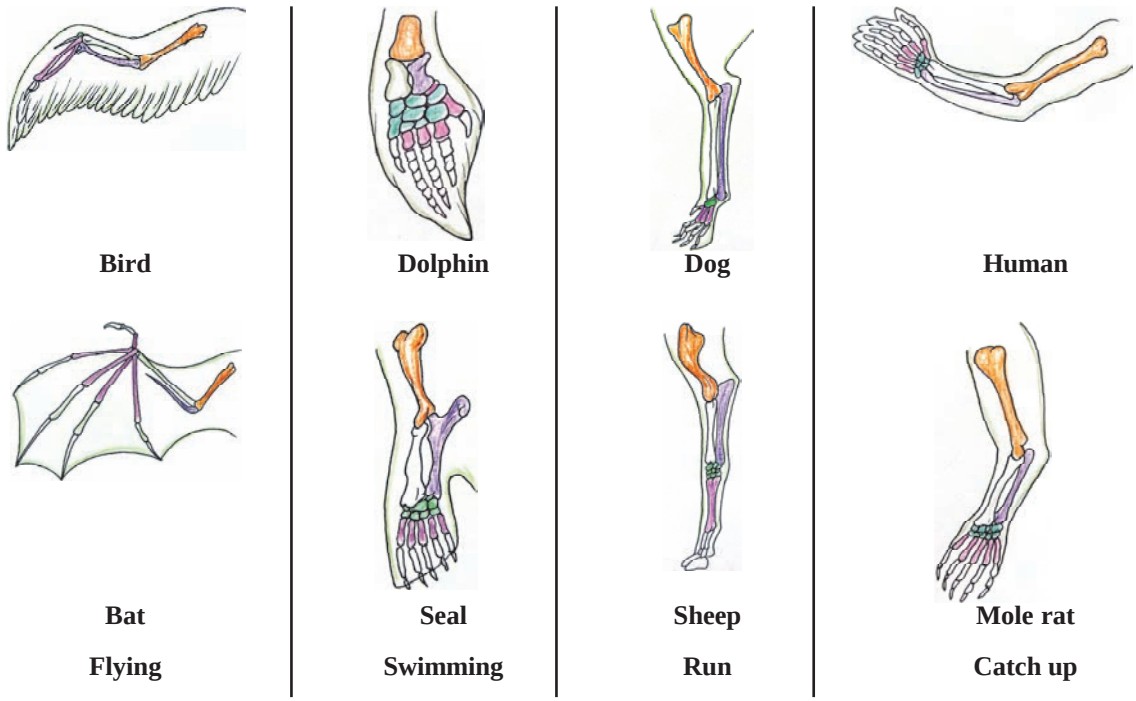


Figure-8(b): Some homologous characters in animals

You can observe the wings of bat, the flippers of dolphins and seal, the forelimb of sheep and dogs as well as humans and mice (in figure 8b) all apparently have the similar set of bones in them. As for example the humerus, radius and ulna are prominently visible in all these organisms. This suggests a common ancestry of these organisms though the appendages are used differently in dolphins, bats, sheep and humans. Characters that indicate similar origin but diverse functions are called as homologous organs. Contrary to this, if we find functional similarity in organs of different origin like the wings of birds and butterflies, we call such organs as 'analogous organs'. While the wings in birds have developed from their forelimbs those of butterflies are mainly from their skin. How could apparently similar organs take up different roles overtime was the question that Darwin and scientists even before Darwin thought about?

1.4.2 Selection and Evolution

As the journey of the Beagle came to an end and the crew returned to England, Darwin started putting together pieces of information that he had collected all through the voyage from the different continents. He started writing an account on his observations. He was aware of the fact that humans could bring about changes in characters in domesticated animals or in crop plants by selecting organisms with preferable characters and breeding them. Selective breeding among a species of organisms with preferable characters by humans, is called as, 'artificial selection'. Darwin started thinking that if humans could select and bring about changes in characters of organisms, could the presence of diverse organisms in nature be a result of some process of selection occurring there?

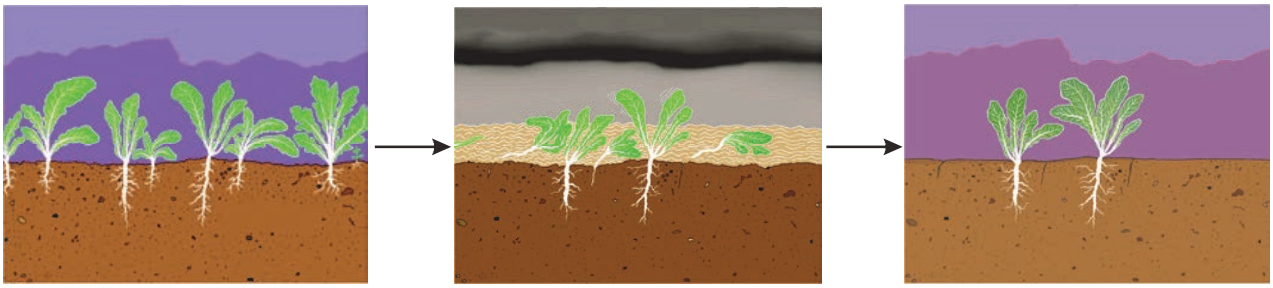
- What type of selection is selective breeding by humans called?

Darwin suggested that a process of natural selection was instrumental in the evolution of one form to the other, resulting in the diverse forms of species found in nature. To understand the process of natural selection you would need to do the project work on it as given in page number.....

- What do we understand about natural selection by doing the project work?
- Is natural selection an intentional process of selection by nature as in artificial selection?

Activity-2

A few figures have been given here. Observe them and suggest the type of selection process shown here. Justify your answer.



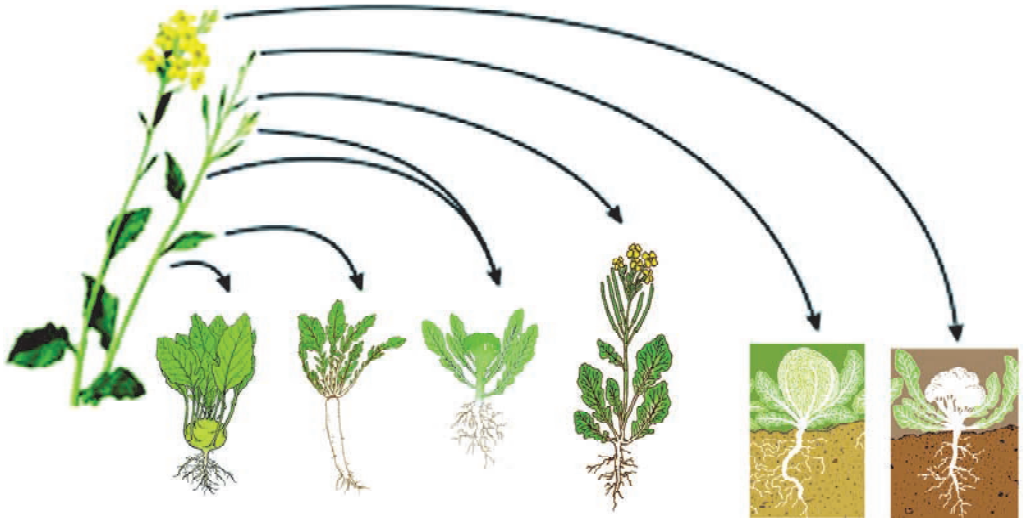
Radish plants growing on a field | radish plants swept away by floods | radish plnts left on the field.

Figure-9: Effect of flood on a radish field (visit site http://www.khaydock.com/books.php for details.)

Nearly two decades after the voyage of the Beagle, another scientist named Alfred Russell Wallace set out on similar voyages from England mainly to islands in the American and Asian subcontinents. He had mainly done an extensive study on the butterflies and mammals of these regions and come to similar conclusions as Darwin. Wallace also suggested that, organisms evolved from preexisting organisms and selective processes in nature occurred spontaneously and non-intentionally giving rise to the diverse forms of species present on earth.

Do you know?

Artificial selection has led to the evolution of different varieties of rice, wheat, etc. that we see today. Different types of vegetables and fruits have also been developed in this way.



Different varieties of mustard Parts preferably consumed by humans | Kholrabi stem | Radish root | Broccoli flower bud | Mustard seeds leaf | Cabbage leaf bud | Cauliflower flower bud

1.4.3 Theory of Evolution

Darwin and Wallace together suggested the following based on evidence and rigorous study-

1. Variations found in organisms led to the diversity in the ability of them to survive in nature. Some live long while others for a short period of time. Some produce more off springs while others produce less off springs. Populations of those that have greater ability to survive, have more chances of evolving over the others. Thus, over time we have new composition of the population of organisms with those apparently suited better to certain sets of environmental conditions.
2. New varieties of organisms may emerge suddenly from preexisting one due to natural calamities.
3. Various organisms have had similar ancestors and thus the evolutionary relationship may be represented by a diagram showing a branched tree(see figure 10).
4. Variations are found in populations of organisms. Only some variations or variable characters pass on from one generation to the next.

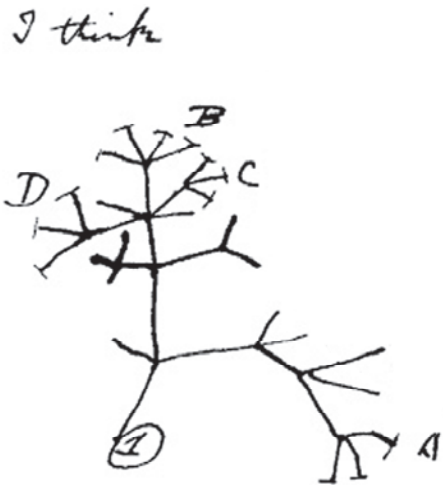


Figure-10: Diagram showing evolutionary relationship as drawn by Darwin

Such variations over generations lead to formation of populations of organisms that may show remarkable changes from their ancestral types. These variations often prove to be advantageous for the organisms that have them over others that do not have them.

6. Populations of organisms that have advantageous variations usually start growing more profusely in an area, affecting the population composition there. Those with less advantageous variations are nearly replaced by those that bear more advantageous variations leading to evolution.

Do you know?



Branched tree representation of Darwin in the context of 5 kingdom classification

- Could the extinction of dinosaurs be due to the evolution of population of organisms having more advantageous variations? What is your opinion about this?

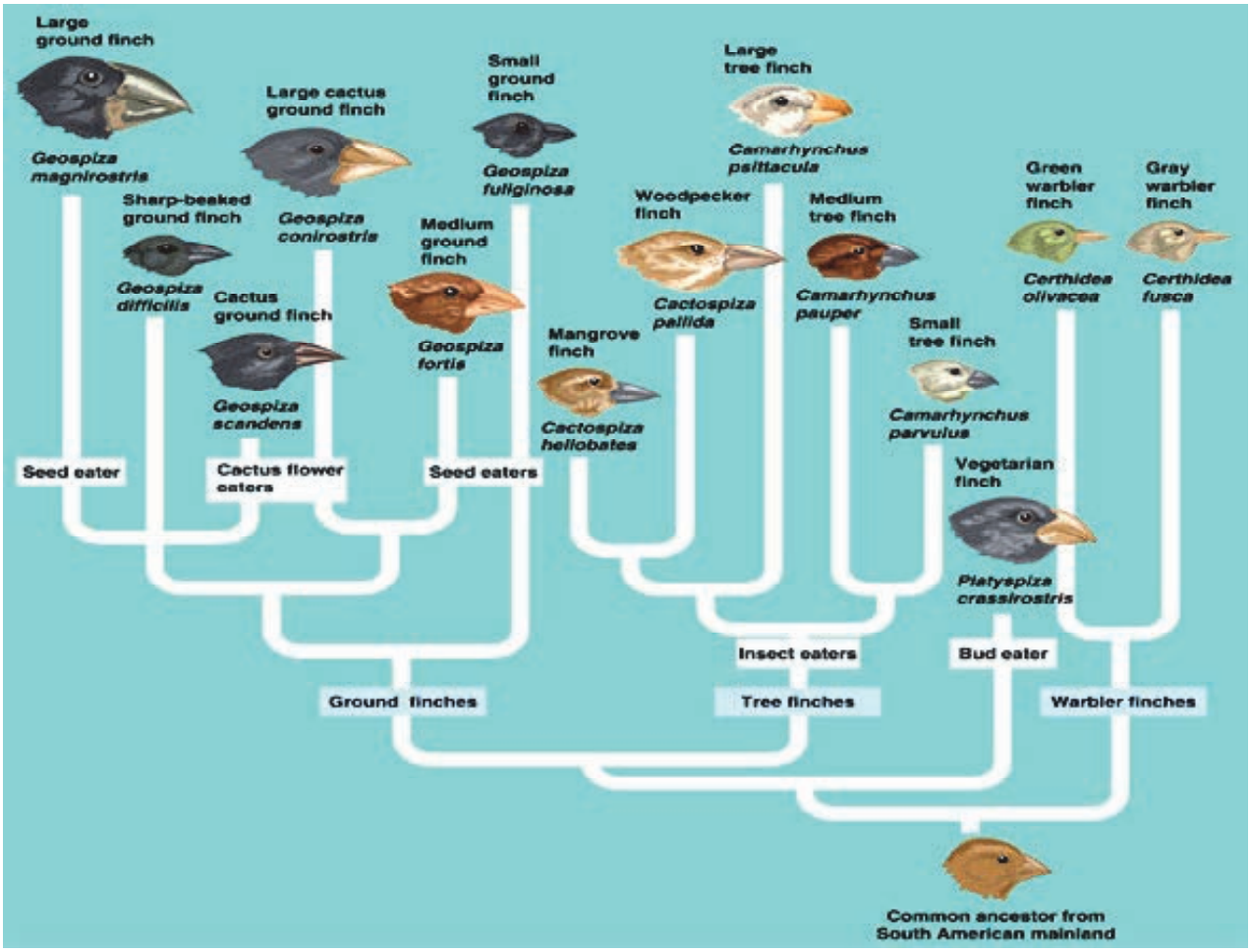


Figure-11: Branched tree of evolution of finches showing common ancestry

Do you know?

Darwin had written the book "On the origin of species by natural selection" while Wallace the book "Darwinism" in which they proposed the evolutionary theory and provided evidences for the same. These books were published in the year 1859 and 1889 respectively

1.5 Process of speciation on the basis of Theory of Evolution

1.5.1 Adaptation and speciation

We have studied that the beaks of finches showed variations according to the availability of food resources. Populations of finches had flown from Ecuador and according to availability of resources different population of finches became suited to environment of different islands. The process of change by which organisms or species become better suited to their environment is called as adaptation.

Were dinosaurs not well adapted to the changing conditions of environment that they could not survive and became extinct?

- How do you think the fishes of Kotumsar caves adapt to the environment of the cave?

Darwin had observed that the finches of Galapagos were so varied that finches of one island could not be bred with finches of other islands. The evolution of organisms such that they vary so much as not being able to interbreed has been considered as the process of speciation. The process may fast due to natural calamities or slow otherwise.

1.5.2 What are species?

Species were considered to be those groups of organisms even during Darwin's time that could sexually interbreed and produce offsprings that were viable(meaning here is -having the ability to reproduce). The definition of species on the basis of their ability to interbreed sexually is considered as 'biological species concept'.

Darwin was concerned regarding this definition as he had observed several species of tortoise, reptiles etc. that could interbreed but were so varied that it would have been better to consider them as a new species. Though the definition of biological species has helped us to organize them under the schema of the systems of classification from two kingdom to five kingdom etc. it has not helped us in organizing groups of organisms like bacteria, certain plants etc. that reproduce asexually.

Darwin had been conducting experiments to find out how variations came to occur among organisms and how they were inherited. He had conducted extensive experiments on plants like Mirabilis(the nine o clock plant) and animals like pigeons. Quite unknown to him a contemporary of Darwin, Gregor Johann Mendel had actually described the process by extensive experiments that we shall study about in chapter 15 (Heredity).

Key words

Evolution of organisms, speciation, adaptation, natural selection, artificial selection, fossils, homologous characters, analogous characters.



What we have learnt?

1. According to the theory of evolution, variations in populations of organisms in nature lead to changes in composition of populations over time. Thus a new population of species emerges from an old one by the process of natural selection.
2. A lot of effort is put into collecting facts and evidences and test them before proposing a theory.
3. Artificial selection is the process of selection of desirable characters of organisms by humans.
4. The process of selection occurring in nature without human intervention, by changes in mainly environmental conditions is called as natural selection.
5. Adaptation is the process of change by which organisms or species become better suited to their environment.
6. Groups of organisms that can interbreed sexually producing viable (capable of further reproduction) off-springs are called as species.This definition of species on the basis of their ability to interbreed sexually is considered as 'biological species concept'.

Exercise

1. Choose the correct answer-

   (i) Name the process through which modern breed of pet dog was acquired-

   (a) Natural selection (c) Juvenile selection

   (b) Artificial selection (d) Work selection

   (ii) Forelimbs of dog and sheep are for walking, whale for swimming and that of bat for flying. All these are examples of-

   (a) Analogous (c) Undeveloped organ

   (b) Homologous (d) all above

(ii) Which of the following are analogous structures-

(a) Wings of bat and butterfly (b) Gills of prawn and fish

(c) Thorns of bougainvillea and tendrils of bottle gourd

(d) Wings of bat and legs of horse

(iii) While proposing his theory, Darwin was influenced by-

(a) The observations of his voyage (b) Cell theory

(c) Law of genetics of Mendel (d) None of the above

2. Fill in the blanks

(i) A group of organisms which can interbreed sexually producing viable off-springs is known as .....

(ii) Difference among beaks of finches show the ..... among them.

(iii) Some bacteria can grow in the medium containing streptomycin (antibiotic). The reason behind this is .....

3. There are different types of bacteria growing in our intestines. They are capable of reproducing mainly asexually(without involvement of sex cells) very fast about once every 20 minute. During reproduction variations emerge and some of these variations are heritable, that is, they pass from one generation to the next. Due to this, we find variations in populations of bacteria growing in our intestines. This is how they evolve very fast. We often consume antibiotics that kill most of these bacteria but some always remain on which the antibiotics may have no effect. This is simply because of the variations present among the bacteria. Bacterial types on which antibiotics have no effect are said to be antibiotic resistant.

Answer the following on the basis of this information.

(a) Justify that natural selection is responsible for evolution among bacterial population.

(b) During which process does variation arise in bacteria?

(c) What is the role of variation in the process of evolution? Use the example of reproduction of bacterial population to elaborate your answer.

(d) "Evolution can be slow or very fast" .Give an example each of slow and fast process of evolution.

(e) Does a better adapted population of bacteria become antibiotic resistant? Give reasons for your answer.

4. Write two difference between natural and artificial selection.

5. What do we come to know from the theory of evolution as proposed by Darwin and Wallace?

6. What are the main points of the theory of evolution of organisms?

7. What is the role of selection and adaptation in the process of evolution?

8. Observe the figure and say whether it represents homologous or analogous structures? Justify your answer.



leaf of pitcher plant | tendril of pea | bract of bougainvillea | spines of cactus

9. (i) Observe the branched tree of evolution given in the figure and find out the closest relative of humans.

(ii) Write the names of two organisms, by observing the figure,that show maximum similarity. Also write the names of two organisms that show maximum differences.

(iii) What do you understand from the figure about evolution of organisms shown there? Write a brief description to elaborate your answer.

